Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A thin rectangular magnet suspended freely has a time period of oscillation equal to T. Now it is broken into two equal halves (each having half the original length) and one piece is made to oscillate in the same field. If its period of oscillation is T', the ratio T'/T is

(A) $\frac{1}{2\sqrt{2}}$

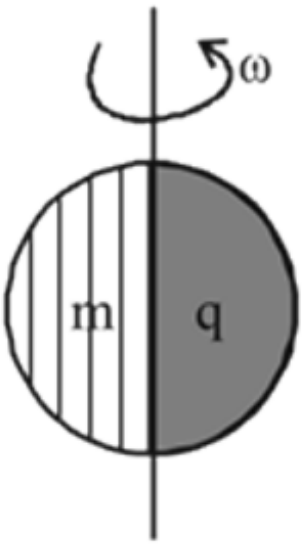
(B) $\frac{1}{2}$

(C) 2

(D) $\frac{1}{4}$

Q.2

Consider a hypothetical spherical body. The body is cut into two parts about the diameter. One of The body is rotated about the axis has identical charge distribution q. hemispherical portion has mass distribution m while the other portion has identical charge distributionq. The body is rotated about the axis with constant speed ω. Then, the ratio of magnetic moment to angular momentum is



(A) $\frac{\mathrm{q}}{2\,\mathrm{m}}$

(B) $> \frac{\mathrm{q}}{2\,\mathrm{m}}$

(C) $< \frac{\mathrm{q}}{2\,\mathrm{m}}$

(D) cannot be calculated

Q.3

Two short bar magnets of magnetic moments M each are arranged at the opposite corners of a square of side d

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such that their centres coincide with the corners and their axes are parallel. If the like poles are in the same direction, the magnetic induction at any of the other corners of the square is

(A) $\frac{\mu_0}{4\pi}\frac{M}{d^3}$

(B) $\frac{\mu_0}{4\pi}\frac{2M}{d^3}$

(C) $\frac{\mu_a}{4\pi}\frac{M}{2d^3}$

(D) $\frac{\mu_0}{4\pi}\frac{M^2}{2d^3}$

Q.4

If the angular momentum of an electron is $\vec{J}$
then the magnitude of the magnetic moment will be

(A) $\frac{eJ}{m}$

(B) $\frac{eJ}{2m}$

(C) $eJ2m$

(D) $\frac{2m}{eJ}$

Q.5

Time periods of vibration of two bar magnets in sum and difference positions are 4sec and 6sec respectively. The ratio of their magnetic moments $\mathrm{M_1M_2}$ is

(A) $6:4$

(B) $30:16$

(C) 2.6:1

(D) 1.5:1

Q.6

A short bar magnet is placed in the magnetic meridian of the earth with north pole pointing north. Neutral points are found at a distance of $30\ \text{cm}$ from the magnet on the East - West line, drawn through the middle point of the magnet. The magnetic moment of the magnet in $\mathrm{Am^2}$ is
close to (Given $\frac{mu_0}{4pi} = 10^{-7}$ in SI units and $\mathrm{B_H}$ = Horizontal component of earth's magnetic field $= 3.6 \times 10^{-5}$ tesla)

(A) 14.6

(B) 19.4

(C) 9.7

(D) 4.9

Q.7

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Two short magnets with their axes horizontal and perpendicular to the magnetic maridian are placed with their centres $40\ \text{cm}$ east and $50\ \text{cm}$ west of magnetic needle. If the needle remains undeflected, the ratio of their magnetic moments M_1 and M_2 is

(A) $4:5$

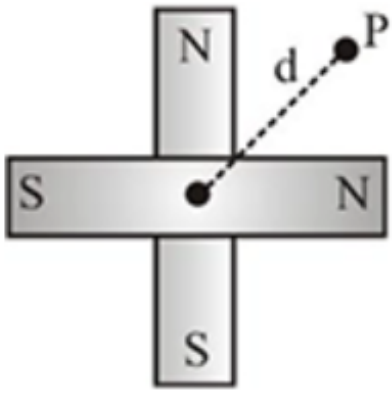
(B) $16:25$

(C) $64:125$

(D) $2:\sqrt{5}$

Q.8

Two short magnets of equal dipole moments M are fastened perpendicularly at their centre (figure)
The magnitude of the magnetic field at a distance d from the centre on the bisector of the right angle is



(A) $\frac{\mu_0}{4\pi}\frac{M}{d^3}$

(B) $\frac{\mu_0}{4\pi}\frac{M\sqrt{2}}{d^3}$

(C) $\frac{\mu_0}{4\pi}\frac{2\sqrt{2}M}{d^3}$

(D) $\frac{\mu_0}{4\pi}\frac{2M}{d^3}$

Q.9

An alternating electric field, of frequency v, is applied across the dees (radius $= R$) of a cyclotron that is being used to accelerate protons (mass $= m$). The operating magnetic field
(B) used in the cyclotron and the kinetic energy (K) of the proton beam, produced by it, are given by:

(A) $B = \frac{mv}{e}$ and $K = 2m\pi^2v^2R^2$

(B) $B = \frac{2\pi mv}{e}$ and $K = m^2\pi vR^2$

(C) $B = \frac{2\pi mv}{e}$ and $K = 2m\pi^2v^2R^2$

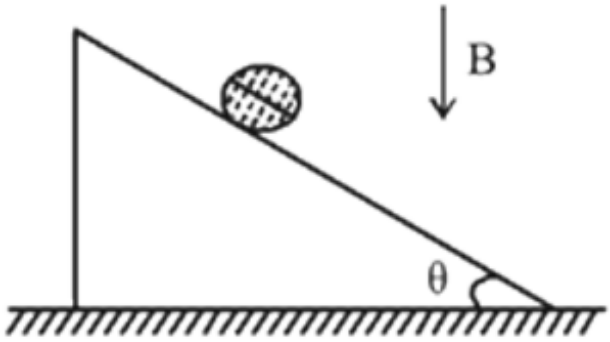
(D) $B = \frac{mv}{e}$ and $K = m^2\pi vR^2$

Q.10

In the figure shown a coil of single turn is wound on
a sphere of radius R and mass M. The plane of the coil is parallel to the plane and lies in the equatorial plane of

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the sphere. Current in the coil is i. The value of B if the sphere in equilibrium is



(A) $\frac{mg\cos\theta}{\pi iR}$

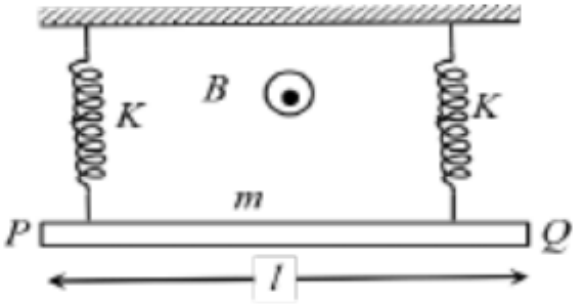
(B) $\frac{mg}{\pi iR}$

(C) $\frac{mg\tan\theta}{\pi iR}$

(D) $\frac{mg\sin\theta}{\pi iR}$

Q.11

A conducting rod of mass m and length 1 is connected by two identical springs as shown in the figure. Initially the system is in equilibrium. A uniform magnetic field of magnitude B directed perpendicular to the plane of the paper outwards also exists in the region. If a curent I is switched on that passes from P to Q through the rod. Further maximum elongation in the spring is [Given: $\mathrm{mg}=\mathrm{Bl}$]



(A) $\frac{BIl}{K}$

(B) $\frac{BIl}{4K}$

(C) $\frac{BIl}{8K}$

(D) $\frac{BIl}{16K}$

Q.12

A conductor in the form of a right angle ABC with $\mathrm{AB}=3\ \mathrm{cm}$ and $\mathrm{BC}=4\ \mathrm{cm}$ carries a current of $10\ \mathrm{A}$. There is a uniform magnetic field of $5\ \mathrm{T}$ perpendicular to the plane of the conductor. The force on the conductor will be

(A) $1.5\ \mathrm{N}$

(B) $2.0\ \mathrm{N}$

(C) $2.5\ \mathrm{N}$

(D) $3.5\ \mathrm{N}$

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Q.13

A straight wire of mass 200gm and length $1.5\ \text{m}$ carries a current of $2\ \text{A}$. It is suspended in mid-air by a uniform horizontal magnetic field B. The magnitude of B (in tesla) is $\left(g = 9.8\ \text{m/s}^2\right)$

(A) 2

(B) 1.5

(C) 0.55

(D) 0.66

Q.14

Three long straight wires are connected parallel to each other across a battery of negligible internal resistance. The ratio of their resistances are $3:4:5$. What is the ratio of distances of middle wire from the others if the net force experienced by it is zero

(A) $4:3$

(B) $3:1$

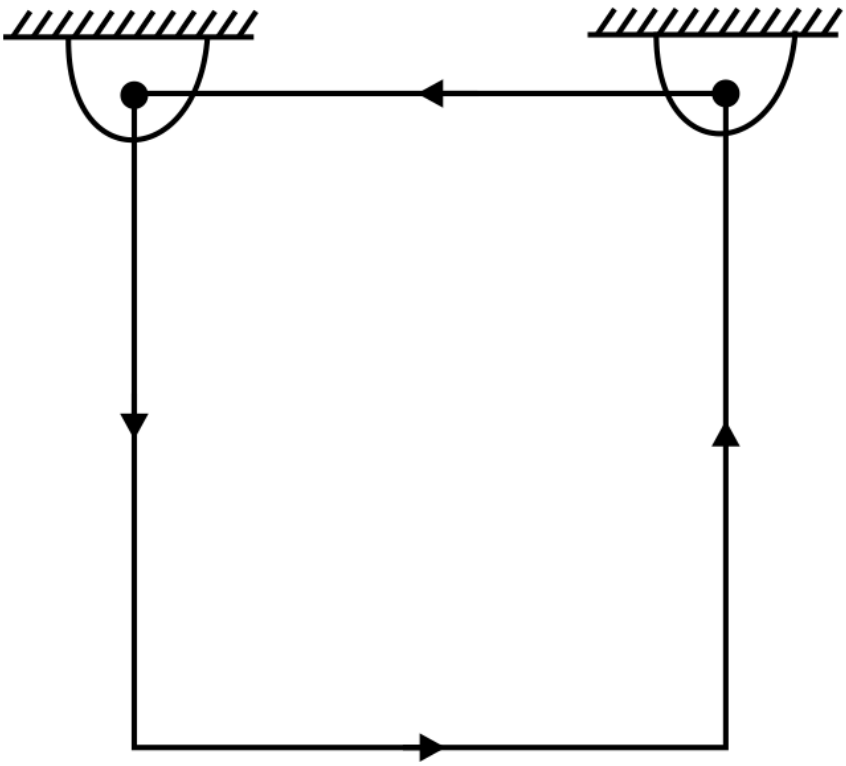
(C) $5:3$

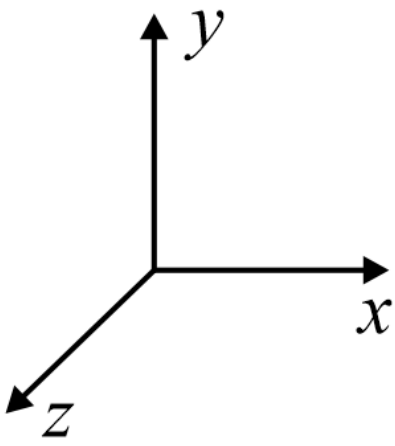
(D) $2:3$

Q.15

A current-carrying uniform square frame is suspended from hinged supports as shown in the diagram such that it can freely rotate about its upper side. The length and mass of each side of the frame are 2 m and 4 kg respectively. A uniform magnetic field $\vec{B} = 3\hat{i} + 4\hat{j}$ is applied. When the wireframe is rotated to $45°$ from vertical and released it remains in equilibrium. What is the magnitude of current (in A) in the wireframe?

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(A) 12

(B) 14

(C) 16

(D) 10

Answers & Solutions

Q.1 Answer:

$\frac{1}{2}$

Solution:

$T = 2\pi\sqrt{\frac{I}{MB}}$

where $I =$ moment of inertia of the magnet about the axis of rotation which is

$I = \frac{mL^2}{12}; \quad L =$ length of magnet,

$m =$ **mass of magnet** $M =$ **magnetic moment** $= pL (p =$ **pole strength) and** B **is the magnetic field. If the magnet is broken into two parts, each of length** $L/2$**, the pole strength** p **remains the same. The magnetic moment of each piece is**

$M' = p \times \frac{L}{2} = \frac{M}{2}$

Since the mass of each piece is $m' = \frac{M}{2}$**, and length** $L' = \frac{L}{2}$**, the moment of inertia of a piece is**

$I' = \frac{m'(L')^2}{12} = \frac{\frac{m}{2} \times \left(\frac{L}{2}\right)^2}{12} = \frac{mL^2}{8 \times 12} = \frac{I}{8}$

Time period T' **is**

$T' = 2\pi\sqrt{\frac{I'}{M'B}} = 2\pi\sqrt{\frac{I/8}{BM/2}} = \frac{2\pi}{2}\sqrt{\frac{I}{MB}}$ **(2)**

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Dividing (2) by (1)

$\frac{T'}{T} = \frac{1}{2}$

Q.2 Answer:

$\frac{\mathrm{q}}{2\,\mathrm{m}}$

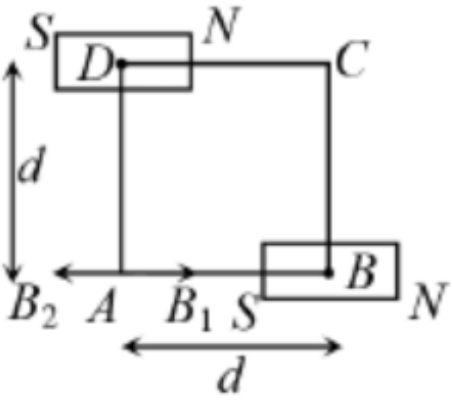
Solution:

The ratio $\mathrm{M/L}$ is always $\frac{\mathrm{q}}{2\,\mathrm{m}}$

Q.3 Answer:

$\frac{\mu_0}{4\pi}\frac{M}{d^3}$

Solution:



Magnetic induction at point A due to magnet at B (ax1al point)

$B_1 = \frac{\mu_0}{4\pi}\frac{2M}{d^3}$

(along AB)

Magnetic induction at point A due to magnet at D (equatorial point)

$B_2 = \frac{\mu_0}{4\pi}\frac{M}{d^3}$

(along BA)

Resultant $B = B_1 - B_2 = \frac{\mu_0}{4\pi}\left(\frac{M}{d^3}\right)$

Q.4 Answer:

$\frac{eJ}{2m}$

Solution:

We know that,

$\frac{M}{J} = \frac{q}{2m} \quad \therefore M = J \times \frac{e}{2m}.$

Q.5 Answer:

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2.6:1

Solution:

$\frac{M_1}{M_2}=\frac{T_2^2+T_1^2}{T_2^2-T_1^2}=\frac{6^2+4^2}{6^2-4^2}=\frac{52}{20}=(2.6):1$

Q.6 Answer:

9.7

Solution:

Here, $r=30\ cm=0.3\ cm$

we know $\frac{\mu_0 M}{4\pi r^3}=B_H=3.6\times10^{-5}$

$\Rightarrow M=\frac{3.6\times10^{-5}}{10^{-7}}(0.3)^3=9.7 Am^2$

Q.7 Answer:

64 : 125

Solution:

For null deflection $\frac{M_1}{M_2}=\left(\frac{d_1}{d_2}\right)^3=\left(\frac{40}{50}\right)^3=\frac{64}{125}$

Q.8 Answer:

$\frac{\mu_0}{4\pi}\frac{2\sqrt{2}M}{d^3}$

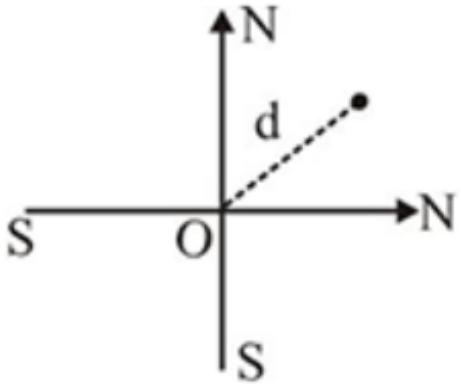
Solution:

Resultant magnetic moment of the two magnets is

$M_{nx}=\sqrt{M^2+M^2}=\sqrt{2M}$

Imagine a short magnet lying along OP with magnetic moment equal to $M\sqrt{2}$**. Thus point** P **lies on the axial line of the magnet**

$\therefore$ **Magnitude of magnetic field at** P **is given by** $B=\frac{\mu_0}{4\pi}\cdot\frac{2\sqrt{2}M}{d^3}$



Q.9 Answer:

$B=\frac{2\pi mv}{e}$ and $K=2m\pi^2v^2R^2$

Solution:

Time period of cyclotron is

$T=\frac{1}{v}=\frac{2\pi m}{eB};B=\frac{2\pi m}{e}v;R=\frac{mv}{eB}=\frac{p}{eB}$

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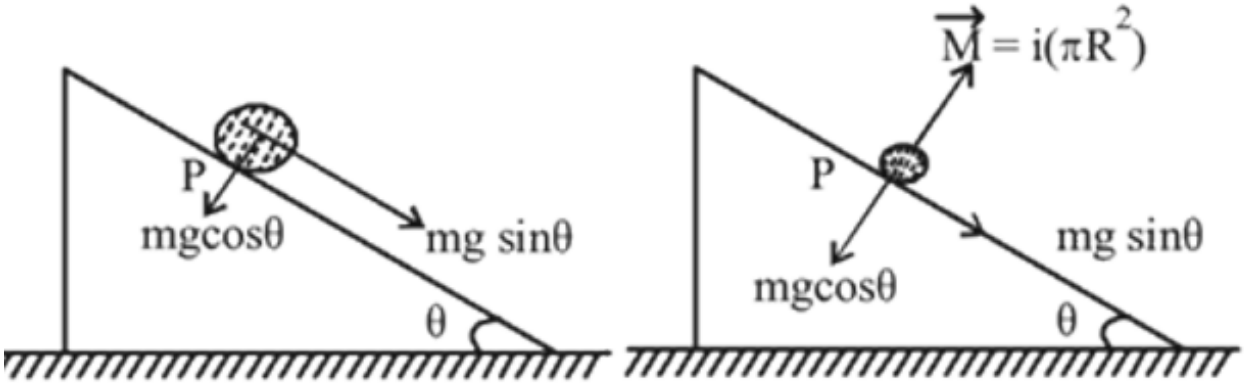
$\Rightarrow \quad p = eBR = e \times \frac{2\pi mv}{e} R = 2\pi m \cup R$

K.E. $= \frac{p^2}{2m} = \frac{(2\pi m \cup R)^2}{2m} = 2\pi^2 mv^2 R^2$

Q.10 Answer:

$\frac{mg}{\pi iR}$

Solution:



For rotational equilibrium of the sphere about point P. $(mg \sin\theta)R = M(B \sin\theta)$

or, $mgR \sin\theta = i\pi \mathrm{R}^2 B \sin\theta$

$B = mg/\pi iR$

Q.11 Answer:

$\frac{BIl}{K}$

Solution:

By work energy theorem

$mgz + Bhz - \int 2k(x+z)dz = 0$

$2mgz = 2k\left[\int_0^2 xdz + \int_0^1 zdz\right]$ **(where x is elongation in the equilibrium position)**

$2mg = mg + kz$

$z = \frac{mg}{k} = \frac{Bl}{k}$

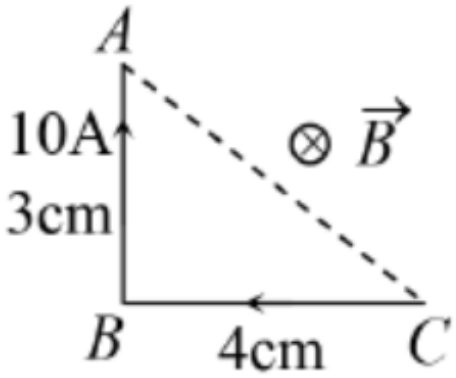
$\therefore$ **(a)**

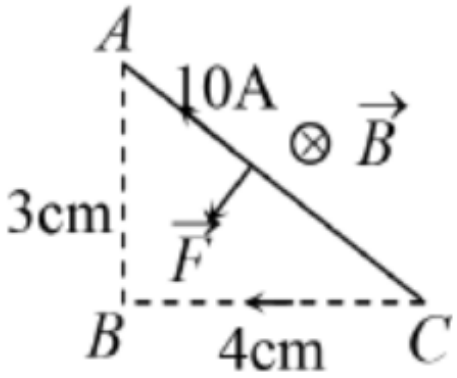
Q.12 Answer:

2.5 N

Solution:

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Force on the conductor ABC
$=$ **Force on the conductor** AC
$= 5 \times 10 \times \left(5 \times 10^{-2}\right) = 2.5\ \mathrm{N}$
$\therefore$ **(c)**

Q.13 Answer:

0.66

Solution:

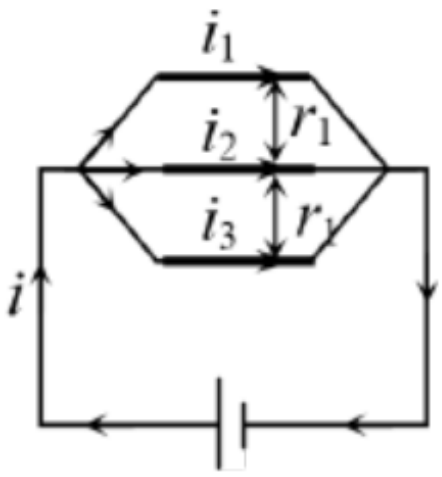
$B = \frac{mg}{\Pi} = 0.66\ \mathrm{T}$

(d)

Q.14 Answer:

$5 : 3$

Solution:



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$i_1 = \frac{k}{3}; i_2 = \frac{k}{4}; i_3 = \frac{k}{5}$

$\frac{\mu_0 i_1 i_2}{2\pi r_1} = \frac{\mu_0 i_2 i_3}{2\pi r_2}$

$\therefore \quad \frac{r_1}{r_2} = \frac{i_1}{i_3} = \frac{5}{3}$

$\therefore$ **(c)**

Q.15 Answer:

10

Solution:

$\vec{\mu}$ **(Magnetic moment of loop) when it is lifted by** $45° = il^2 \frac{\hat{j} + \hat{k}}{\sqrt{2}}$

$\therefore$ $\vec{\tau}$ **due to magnetic field** $= \vec{\mu} \times \vec{B} = \frac{il^2}{\sqrt{2}} \hat{j} + \hat{k} \times 3\hat{i} + 4\hat{j}$

$\vec{\tau}$ **due to** mg **(about top edge)** $= 4mg\frac{l}{2}\cos 45° \quad \hat{i}$

$\therefore$ **For equilibrium net torque along** x**-axis** = 0

$\therefore \quad \frac{4mgl}{2\sqrt{2}} = \frac{4il^2}{\sqrt{2}} \Rightarrow i = \frac{mg}{2l} = 10 \quad A$

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